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COMMENTARY

Rethinking clinical trials in the multimorbidity era: the imperative for patient-centered tailored approaches

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Abstract

Background and Objectives: Multimorbidity, the coexistence of two or more chronic conditions, affects about 40% of all adults and over half of adults over 60 years. The complexity of multimorbidity (MM) often renders traditional trial designs inadequate, unable to account for the context of interventions, including the interplay of multiple health conditions in daily life. This gap reduces the generalizability and applicability of their results.

Methods: This commentary aims to review the current state of trials targeting or involving patients with MM. Highlighting current limitations and drawing on insights from an international dedicated workshop in Bielefeld, Germany, we identify an ongoing and pressing need for innovative, patient-centered approaches to their design and conduct.

Results: We propose a shift toward more holistic and integrative experimental approaches, including developing interventions tailored to the characteristics and needs of patients with MM, establishing relevant outcomes, and enhancing data collection and process evaluation. We specifically advocate for adaptive trial designs, prespecified subgroup analyses, and the incorporation of patient-reported outcomes and

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experience measures such as burden of care to ensure that research is both comprehensive and reflective of the needs of patients living with MM, their caregivers, and of the clinicians participating in their care. Ethical considerations are discussed in our commentary as well, emphasizing the importance of patient safety, data protection, and informed consent. Finally, we call for the development of specific reporting guidance, such as a SPIRIT extension tailored to MM trials, to help researchers adapt standard protocols to the complex and heterogeneous reality of this population.

Conclusion: This commentary aims to bridge the gap between research and practice, fostering the development of effective interventions that improve patient outcomes and enhance the quality of care for patients living with MM.

Keywords: Multimorbidity; Patient-centered trials; Adaptative trial design; Patient-reported outcomes; Real world evidence; Complex interventions

1. Introduction

Multimorbidity, defined as the coexistence of two or more chronic conditions [1], has become a critical issue in global health care. It affects approximately 40% of the general population and more than half of adults over 60 years, more than any single health condition globally [2]. Projections indicate a continued increase over the next 20 years, with those having four or more conditions nearly doubling [3]. This places a major burden on patients, caregivers, and health-care systems, which struggle to meet higher demands with services designed for single conditions [4].

Multimorbidity is present in many forms, ranging from combinations of common chronic diseases to complex constellations involving severe physical, mental, or social problems. Its expression and impact vary across care settings—primary, secondary, and community—and are strongly influenced by social disadvantage and mental health comorbidity. This heterogeneity has major implications for research; interventions effective in one context or group may not work in another, and trial designs must reflect this diversity to enhance the validity, applicability, and effectiveness of resulting interventions.

In high-income countries, 80% of primary care consultations involve patients with multimorbidity (MM) [5]. However, current disease-specific treatment guidelines often fail to account for interactions between multiple conditions and their treatments, resulting in suboptimal care [6]. While exclusion or underrepresentation of patients with MM in clinical trials is well-documented [7], the challenge goes beyond inclusion. Trial designs must address the complex syndromes that emerge within MM, delivering robust evidence to support co-created care plans by patients, caregivers, and clinicians.

Traditional trial designs inadequately reflect these realities. Tailored trials that incorporate the interactions between daily life, multiple diseases, and their treatments can produce more reliable estimates of the efficacy, effectiveness, and safety of interventions. This commentary advocates the development of such trials to enhance their generalizability and usefulness. We use the term "patients" for consistency, while being aware that this does not do justice to the multiple roles people with MM play beyond receiving care.

International symposia held in Frankfurt/Main, Germany, in 2012 and Stockholm, Sweden, in 2018 underscored a growing global consensus on improved approaches to MM research. A dedicated workshop held in Bielefeld, Germany, in May 2024 focused on redefining trial methodologies to better capture the intricacies of MM. Twenty-three multidisciplinary experts from four continents—including general practice, geriatrics, clinical pharmacology, internal medicine, health promotion and public health, health psychology, nursing, evidence-based medicine, epidemiology, and statistics—deliberated on how best to adapt research frameworks to the needs of patients with MM. Insights from these discussions form the basis of the following recommendations to make future trials more useful for clinicians, patients, and policymakers.

2. Current state of clinical trials in MM

Recent efforts in conducting clinical trials evaluating interventions aimed to improve care and outcomes for patients living with MM have been primarily driven by the need to identify optimal interventions for patient management. Despite this focus, trials evaluating health service interventions in primary care settings have largely not demonstrated that such interventions improve patient outcomes [8]. Several systematic and scoping reviews have examined the effectiveness of various primary care intervention strategies, ranging from care coordination to support self-management [8–10]. However, these reviews have consistently reported inconclusive results, with significant challenges in patient selection and the practical implementation of interventions [11]. Similarly, other trials investigating multidisciplinary, patient-centered interventions to enhance patient engagement and self-management have failed to demonstrate significant effects compared to usual care [8–10]. Moreover, most initiatives aimed at optimizing medication management suggest improvements in the appropriateness of prescribed medications but fail to show significant differences in patient-important outcomes [12,13].

What is new?

Key findings
- Traditional trial designs are poorly suited to the realities of MM. This paper outlines methodological and ethical priorities for patient-centered, adaptive trial design.

What this adds to what is known?
- Integrates international expert consensus into actionable guidance for MM trials, including adaptive, mixed-methods, and co-designed approaches.

What is the implication and what should change now?
- Researchers should adopt flexible, proportionate, and context-sensitive designs and develop dedicated reporting guidance (eg, a SPIRIT extension) to ensure relevance and transferability.

Trials targeting MM often focus on comorbidity and thus adopt disease-centered methods, focusing on index conditions and overlooking the complex interplay of co-existing conditions and their management. As a result, their generalizability and applicability in clinical settings are limited, and their outcomes, often about individual disease control measures, do not align with patient goals and priorities, rendering them inadequate [8]. Additionally, while there is evidence that interventions targeting comorbid depression improve depression outcomes, there is less evidence addressing the bidirectional impact between mental health disorders and chronic physical conditions, thereby overlooking the need for holistic solutions [9]. Furthermore, evaluation studies of organizational integrated care interventions mostly focus on the effects on service utilization (eg, emergency department visits, hospital admissions, and length hospital of stay), less on costs, and seldom on the patient experience of care itself. In addition, trials addressing clinical interventions on the problematic and complex situation of patients living with MM have provided inconsistent results [9]. The resulting gaps have left clinicians less able to develop plans of care that respond well to the complex situation of each patient.

3. Proposal for future research directions

The research landscape indicates a pressing need for a paradigm shift in how clinical trials are designed and conducted, emphasizing holistic, patient-centered approaches that address the real challenges of MM care. The outcomes of the Bielefeld Workshop Echo International Policy call to advance understanding, prevention, and management of this global issue [4].

This paradigm shift involves both refining existing trial methods and rethinking how evidence is generated, aligning patient-reported outcome measures (PROMs) and patient-reported experience measures (PREMs) with patient priorities and reducing data burden to adopting adaptive, pragmatic, and routine record/registry-based designs that overcome recruitment barriers and better reflect real-world care. Realist and longitudinal approaches further enrich this agenda by explaining how and why interventions work, for whom, and under what circumstances, linking mechanisms to outcomes across time and context [14]. For instance, adaptive or platform trials using routine data and low-burden PROMs/PREMs collected through patient-preferred channels can operationalize these principles. Ethical innovation must accompany these developments, when risks are minimal and interventions reflect usual care, bringing it closer to evidence-based best practice; cluster randomization with simplified or waived consent can enhance inclusion and equity. Pragmatic trials using routine electronic health record data while capturing key PROMs illustrate this balance. Finally, leveraging large-scale health data and machine learning tools can enable adaptive randomization, continuous evaluation, and learning health system models that shift resources toward co-design, implementation, and contextual understanding. For example, real-time analytics in chronic care registries can support adaptive trial designs and rapid feedback loops for quality improvement.

Below, we outline key research directions derived from the workshop discussions on clinical approaches, methodological frameworks, and innovative intervention strategies, including a comparison of selected items from the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) protocol for reporting guidelines [15] with our proposed recommendations (Table 1).

3.1. Addressing methodological challenges in MM trials

Trials involving patients with MM face distinct challenges due to the heterogeneity of conditions, fluctuating disease courses, and complex treatment and health-care service interactions. To produce relevant and actionable evidence, MM trials require flexible, innovative methods.

Prespecified subgroup analyses may also be helpful [16]. They can reveal which populations may benefit most, enabling more personalized care strategies. These analyses should consider not only demographic factors (eg, age, sex, socioeconomic status [SES], and ethnicity) but also the clinical complexity of MM, such as type, severity, and duration of co-existing conditions. To ensure credibility, sub-group analysis findings should be appraised using tools like Instrument to assess the Credibility of Effect Modification Analyses (ICEMAN), which help guard against misleading interpretations [17].

Table 1. Essential actions and considerations necessary for trustworthy, useful, and generalizable trials in effective MM research

SPIRIT checklist item	Explanation of SPIRIT item	Workshop conclusions and actions for multimorbidity research
Trial design	Description of the trial's structure and methodology.	Use adaptive trial designs with oversight from an independent committee. Report adaptations transparently.
Study setting	Description of locations and settings where the trial is conducted.	Select care settings that reflect real-life multimorbidity care, including fragmented services and support networks. Use PROGRESS-Plus to guide equity.
Eligibility criteria	Criteria for patient selection, including inclusion and exclusion factors.	Include diverse patients with cognitive, physical, or social complexity. Consider SES, health literacy, and caregiver support.
Interventions	Detailed information on interventions for each group to allow replication.	Co-design interventions with patients and professionals. Ensure adaptability. Use structured frameworks for description.
Outcomes	Primary and secondary outcomes to be measured, including their relevance and timing.	Prioritize outcomes important to MM patients (function, QoL). Use sensitive PROMs and PREMs. Include process measures. Align with causal pathways. Involve patients. Refine existing COS. In addition, embed qualitative indicators to capture patient and caregiver experiences, contextual factors, and perceived changes in care processes complementing quantitative measures.
Data collection methods	Plans for how and what data will be collected to ensure quality and relevance.	Prioritize meaningful outcomes, even beyond routine data. Reduce burden via PPI. Support diverse needs. Use guided PROMs. Avoid 'scope creeping'. Train staff and adapt. Include process evaluations to assess fidelity and context using GRIPP. Use community-based strategies for inclusive engagement. Combine quantitative and qualitative methods to ensure a comprehensive understanding of intervention mechanisms (theory of change) and contextual influences.
Statistical methods	Plans for analyzing collected data, including any additional or subgroup analyses.	Use prespecified subgroup analyses reflecting clinical and demographic diversity. Assess credibility with tools like ICEMAN.
Harms	Plans for identifying, recording, and addressing adverse events or other unintended effects.	Monitor QoL, function, and prescribing. Use proportional risk management. Ensure data protection, especially with digital tools.
Ethics and dissemination	Approaches for ethical approval, amendments, and sharing results with stakeholders.	Apply proportional oversight. Minimize burden. Share results transparently with patients and stakeholders. Safeguard data privacy.

COS = core outcome set; GRIPP checklist = Guidance for Reporting Involvement of Patients and the Public; ICEMAN tool = Instrument to assess the Credibility of Effect Modification Analyses; MM = multimorbidity; PPI = patient and public involvement; PREM = patient-reported experience measures; PROM = patient-reported outcome measures; QoL = quality of life; SES = socioeconomic status.

Adaptive trial designs are particularly useful, allowing for preplanned modifications to elements like sample size, eligibility criteria, or intervention components based on interim data. This flexibility supports patient-centeredness and real-world relevance. However, adaptations based on limited or unstable data carry risks; guidance from an independent trial steering committee is essential to safeguard scientific rigor.

3.2. *Better consideration of target populations and settings*

To ensure the applicability of trial results to the care of patients with MM, future research must focus on alleviating the ill health and practical problems that emerge from the complex interactions between co-existing conditions and their impact on each patient's cognitive, physical, and social function. These impacts may result from responding to the daily demands of caring for multiple conditions, including accessing and using poorly coordinated and prioritized health-care services and enacting burdensome self-management routines. Research should also explicitly address how inequalities related to SES, health literacy, and health-care access can affect these care dynamics. Selecting diverse patient groups—with a particular focus on targeting those with the potential to benefit from the proposed intervention—can help avoid ceiling effects and improve relevance. In addition, health services research trials should reflect the multidisciplinary nature of real-world multimorbidity care by involving all health-care professionals with relevant backgrounds. These aims can be supported using appropriate guiding frameworks, such as PROGRESS-Plus [18], that identify social factors—eg, place, ethnicity, occupation, gender, and education—influencing health disparities, and capture the crucial roles professional and informal caregivers play in enabling patient's capacity for problem solving, health-care use, and self-management.

3.3. Designing targeted interventions for MM

Designing interventions for patients with MM requires special attention to the heterogeneity of conditions, treatment burden, and often competing or conflicting treatment goals. The design process must be dynamic and collaborative to ensure that resulting approaches to care are useful, usable, and responsive to the lived experiences of patients. Such interventions may range from clinical approaches (eg, medication or therapy for a specific condition) to more complex clinical strategies such as shared decision-making or medicine optimization and to organizational models that reshape care delivery at team or system level. A key component of this process can be the co-design, whereby patients, caregivers, and multidisciplinary professionals jointly and iteratively develop treatment strategies [19]. This approach can help design more feasible, acceptable, and perhaps more sustainable interventions, especially in populations facing fluctuations in health, functional limitations, or coordination challenges.

Evaluation of these interventions may benefit from hybrid effectiveness-implementation trial designs [20], which simultaneously assess clinical outcomes and the feasibility, fidelity, and acceptability of implementation. This dual focus is particularly important in MM, where integration into routine care and flexibility across settings are essential. Hybrid designs are well suited to the complex and unpredictable realities of MM care, where interventions must adapt to individual needs and contextual factors [21].

3.4. Establishing relevant outcomes

The inclusion of PROMs on health and well-being is essential to reflect perspectives of patients on benefits and harms within their biopsychosocial context [22]. PROMs should be carefully selected for their appropriateness and their sensitivity to change, especially among older adults or functionally limited populations, reflect what matters most to patients with MM, and avoid misleading findings [23]. The number and length of psychometric instruments should be balanced against comprehensiveness, as overlapping questionnaires may interfere with response behavior [24]. Standardized short instruments offer advantages, and computerized adaptive testing can further reduce burden and increase response rates [25,26].

In trials involving participants with severe- or high-impact MM, the burden of outcome assessment can hinder meaningful participation, especially among those with limited cognitive, emotional, caregiving, or social resources. Repeated or lengthy questionnaires may cause fatigue, confusion, or distress, reducing follow-up despite continued engagement. Designing proportionate and compassionate follow-up procedures is therefore critical, prioritizing essential outcomes, simplifying data collection, allowing proxy or remote assessments, and monitoring participant burden throughout the study. Such approaches improve data completeness and uphold ethical standards of respect and patient-centeredness.

Selecting outcomes that are both measurable and responsive to change is particularly challenging. People with complex or high-impact MM—those experiencing several interacting chronic conditions that affect daily functioning and quality of life (QoL)—often require several concurrent or sequential interventions over extended periods, and benefits may emerge unevenly across domains such as function, symptoms, psychological well-being, or social participation. Conventional short-term or disease-specific outcomes may underestimate true effects, while global measures of health or well-being can be difficult to meaningfully change. Trials should therefore combine short-term process indicators with long-term, patient-centered outcomes and consider composite or hierarchical endpoints that capture interrelated changes, eg, combining symptom relief and functional recovery or prioritizing survival followed by QoL. Longitudinal or adaptive assessment frameworks can enhance sensitivity to gradual or cumulative improvements while maintaining feasibility.

Where possible, outcomes should be explicitly aligned with the intervention's causal pathways, linking specific components to expected changes in health, behavior, or system-level effects. The well-being and burden on informal caregivers and clinicians should also be assessed to capture the broader impact of interventions. Engaging patients and other interest-holders through patient and public involvement (PPI) improves the relevance and acceptability of selected outcomes [27]. While core outcome sets (COS) for MM exist [28,29], further refinement is needed to enhance their responsiveness and applicability across diverse contexts and health-care systems.

3.5. Strengthening data collection

Efficient and thoughtful data collection is essential in MM trials, particularly given the substantial burden of illness and treatment experienced by patients and their caregivers. To minimize the additional burden of participation, researchers should engage with PPI advisors to explore patients' needs and preferences and the capability of patients to collect planned data. Where appropriate, routinely collected clinical data can be leveraged to reduce demands on patients, caregivers, and clinicians. In addition, the use of acceptable wearables may be taken into consideration, as a growing number of devices are available which may provide objective data (and reduce recall or measurement biases), eg, on physical function and reduce the burden of long questionnaires and interviews on study patients.

Personalized support should be provided for patients who face challenges related to health literacy, cognitive disabilities, or language barriers. This includes using language-concordant materials, simplified explanations, visual aids, and guided interviews to ensure that all patients can engage meaningfully with data collection processes.

Data collection procedures should be carefully designed to minimize 'scope creep'—the unnecessary expansion of questionnaires or data fields—by focusing only on data, ie, relevant, interpretable, and actionable. Finally, training health-care providers in the ethical, technical, and logistical aspects of data collection, along with ongoing adaptation of data strategies based on patient and clinician feedback, helps ensure both methodological quality and patient-centeredness.

3.6. Strengthening process evaluations

Detailed documentation and analysis of care processes—particularly the dynamics of encounters and the operational aspects of implementation—offer essential insights that enhance the interpretation, transferability, and scalability of interventions. These include assessing to the original design and documenting adaptations made in response to local settings or patient needs.

Qualitative data collection plays a central role in understanding how interventions achieve or fail to achieve their intended effects in real-world contexts. Implementers should articulate the underlying theory of change and use methods such as interviews, focus groups, or ethnographic observation to capture patients' and clinicians' experiences, perceived barriers, and contextual influences often missed by quantitative outcomes. Integrating qualitative and quantitative evidence within mixed methods designs enhances interpretability, supports transferability across settings, and strengthens methodological quality.

Examining how an intervention is implemented within the study setting may reveal effect modifiers, illuminate barriers and facilitators to future implementation, and help differentiate core components from those that need further development. Although the workshop was designed mainly to address quantitative comparability across trials, participants also emphasized the need to embed qualitative approaches to achieve truly patient-centered and context-sensitive evaluations of complex interventions.

The context in which MM interventions are implemented is a decisive determinant of their effectiveness [9]. Organizational and clinical strategies that appear promising in principle may fail when introduced into systems that are fragmented, under-resourced, or poorly coordinated. Improvements in patient outcomes often depend on multiple actors across settings—general practitioners, hospital specialists, family/caregiver, and social and voluntary services—working in concert. Consequently, process evaluations should systematically capture contextual factors such as leadership, communication structures, team capacity, and organizational culture to explain why an intervention succeeds or falters. Many apparent "negative" trial results may reflect a mismatch between intervention strength, system readiness, choice of outcome measures, and outcome sensitivity rather than true ineffectiveness. Monitoring and reporting these contextual dependencies is essential for interpreting trial results and designing interventions that are both feasible and scalable.

PPI is especially valuable in designing process evaluation, ensuring that care experiences of patients with MM—across different age, ethnicity, and socioeconomic contexts—are meaningfully captured. To promote this inclusivity, researchers may adopt community-based engagement strategies. Using standardized tools such as the Guidance for Reporting Involvement of Patients and the Public (GRIPP) checklist [30] helps structure and report PPI activities consistently while enhancing transparency.

3.7. Ethical challenges in MM trials

Multimorbidity trials raise specific ethical considerations due to the complexity of patients' conditions and the cumulative treatment burden they often face. While most interventions in these trials are low-risk—typically aiming to optimize care coordination, prescribing practices, or patient engagement rather than testing new drugs or devices—there remains an ethical obligation to ensure patient safety, data protection, and responsible risk management [9].

Even in low-risk trials, ethical oversight should include careful monitoring of the intervention's effects on patients' QoL, symptom burden, and functional status using appropriate PROMs. Attention should also be given to potential unintended consequences of interventions, such as shifts in prescribing behaviors or the introduction of care routines that could increase workload or complexity.

Risk management procedures should be proportional to the level of risk. In most MM trials, this may mean embedding light-touch monitoring mechanisms into routine care workflows, particularly when interventions support rather than replace usual care (eg, tools for structured medicines review in general practice). More intensive interim monitoring may only be necessary in trials involving substantial changes to care delivery or data systems.

Finally, the use of digital tools, electronic records, or cross-system data integration in MM trials requires strict adherence to local and international data protection standards, ensuring that patient privacy and confidentiality are maintained throughout the study.

4. Conclusion

There is an urgent need for immediate, actionable guidance for research in MM populations. It is imperative that we advance the development of expert methods and incorporate these advancements into formal criteria supporting the design and reporting of trials, such as SPIRIT. By embracing and implementing the proposed research directions, we expect future trials in patients living with MM to generate valid and applicable estimates of effectiveness in real-world settings. Their translation into clinical care should then contribute to enhancing the quality and experience of care for individuals with MM, leading to broader systemic improvements in health-care delivery and patient outcomes.

CRediT authorship contribution statement

Ana Isabel González-González: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Elisa Fabbri:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Lara Schürmann:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Jeanet Blom:** Writing – review & editing, Supervision, Methodology, Investigation. **Svetlana Puzhko:** Writing – review & editing. **Amaia Calderón-Larrañaga:** Writing – review & editing. **Nina Grede:** Writing – review & editing. **Bruce Guthrie:** Writing – review & editing. **María Hanf:** Writing – review & editing. **Gabriele Meyer:** Writing – review & editing. **Rafael Perera:** Writing – review & editing. **Mieke Rijken:** Writing – review & editing. **Martin Scherer:** Writing – review & editing. **Susan M. Smith:** Writing – review & editing, Conceptualization. **Michael A. Steinman:** Writing – review & editing, Conceptualization. **Sharon Straus:** Writing – review & editing, Conceptualization. **José María Valderas:** Writing – review & editing. **Marjan van den Akker:** Writing – review & editing. **Victor M. Montori:** Writing – review & editing, Supervision, Conceptualization. **Christiane Muth:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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